Chapter 19 Notes

Section 2

Notes

For the model in this section to work, we need $\omega_2 \in M$. However, it is not clear to my why this is.

If we assume that $\omega_2 \in M$, here is why $\langle \mathcal{G}, \mathcal{R} \rangle$ is an M-set. It is easy to see that each particular forcing condition is in M - each particular one can be explicitly defined, as it is assumed that $p(\alpha) = \langle \emptyset, \emptyset \rangle$ for all but a finite number of $\alpha \in \omega_2$. The set $\mathcal{G}$ of all forcing conditions is a subset of $\mathcal{P}(\mathcal{P}(\omega_2 \times (\mathcal{P}(\omega \times \omega) \cap M)) \cap M) \cap M$, which is in M. $\mathcal{G}$ is also easily seen to be definable over M, and consequently, $\mathcal{G} \in M$. Then $\mathcal{R}$ is also easily seen to be in M, as it is definable over M, and is a subset of $\mathcal{G} \times \mathcal{G}$, which is in M (as we have shown $\mathcal{G}$ to be in M).

Exercise 2.1

This is basically the same proof as that of Theorem 4.1 of Chapter 18.

Proof. We prove that, for each $\alpha \in \omega_2$, $[\![\sim\text{Const}(f_\alpha)]\!]$ is valid in the model M. Suppose, to the contrary, that $p \nvDash [\![\sim\text{Const}(f_\alpha)]\!]$ for some p. Then for some p' with $p\mathcal{R}p'$, $p' \vDash [\![\text{Const}(f_\alpha)]\!]$. By Theorem 3.2 of Chapter 18, it follows that $p \vDash \Diamond [\![f_\alpha \approx \hat{c}]\!]$ for some contrsuctible set $c \in M$. This means that, for some r with $p'\mathcal{R}r$, $r \vDash [\![f_\alpha \approx \hat{c}]\!]$. We derive a contradiction from this. r is some forcing condition, so suppose that $r(\alpha) = \langle P, N \rangle$. There are then two cases.

Case 1) $c \subseteq P$. Because both P and N are finite, we can pick some integer n such that $n \notin P \cup N$ (thus also $n \notin c$). Let $P' = P \cup \{n\}$. Then there is some forcing condition q such that, for all $\beta \in \omega_2$, if $\beta \neq \alpha$, then $q(\beta) = r(\beta)$, and if $\beta = \alpha$, then $q(\beta) = \langle P', N \rangle$. Obviously $r\mathcal{R}q$. Because $c \in P'$, it follows that $q \vDash [\![\hat{n} \in f_\alpha]\!]$. But because $n \notin c$, $q \vDash [\![\sim(\hat{n} \in \hat{c})]\!]$. These two things jointly imply $q \nvDash [\![f_\alpha \approx \hat{c}]\!]$. But also $q \vDash [\![f_\alpha \approx \hat{c}]\!]$, as $r \vDash [\![f_\alpha \approx \hat{c}]\!]$ and $r\mathcal{R}q$. Thus we have our contradiction in this case.

Case 2) $c \nsubseteq P$. Pick some n in c but not in P. Let $N' = N \cup \{n\}$. Then there is some forcing condition q such that, for all $\beta \in \omega_2$, if $\beta \neq \alpha$, then $q(\beta) = r(\beta)$, and if $\beta = \alpha$, then $q(\beta) = \langle P, N' \rangle$. Obviously $r\mathcal{R}q$. Because $n \in c$, it follows that $q \vDash [\![\hat{n} \in \hat{c}]\!]$. But because $n \in N'$, it follows that $q \vDash [\![\sim(\hat{n} \in f_\alpha)]\!]$ (Reason: Suppose, for some q' with $q\mathcal{R}q'$, that $q' \vDash \hat{n} \in f_\alpha$. Then for some x, $q' \vDash [\![x \epsilon f_\alpha]\!]$ and $q' \vDash [\![x \approx \hat{n}]\!]$. This means x is $\hat{a}$ for some a, and thus, by Lemma 3.2 of Chapter 17, a and n are the same set. But this means $q' \vDash [\![\hat{n} \epsilon f_\alpha]\!]$, which cannot happen, as $n \in N'$). These two things imply that $q \nvDash [\![f_\alpha \approx \hat{c}]\!]$. But also $q \vDash [\![f_\alpha \approx \hat{c}]\!]$ (for the same reason as in the first case), which is our contradiction.

□

Section 3

Exercise 3.1

We prove that $[\![f : \hat{\omega} \xrightarrow[\text{onto}]{\text{1-1}} \hat{\omega_2}]\!]$ is valid in the described $S4$-regular model described. Consider any artibrary world p. It suffices to show that the following four things hold:

$$(1)\ p \vDash [\![(\forall x)(\forall y)(\forall z)((\langle x, y\rangle \in f \wedge \langle x, z\rangle \in f) \supset y = z)]\!]$$

(this is the translate of the formula saying f is a function)

$$(2)\ p \vDash [\![(\forall x)(\forall y)(\forall z)((\langle x, z\rangle \in f \wedge \langle y, z\rangle \in f) \supset x = y)]\!]$$

(this is the translate of the formula saying that f is 1-1)

$$(3)\ p \vDash [\![(\forall x)(x \in \hat{\omega} \supset (\exists y)(y \in \hat{\omega_2} \wedge \langle x, y\rangle \in f))]\!]$$

(this is the translate of the formula which says that, for each $y \in \omega$, there is some $x \in \omega_2$ such that $f(x) = y$)

$$(4)\ p \vDash [\![(\forall x)(x \in \hat{\omega_2} \supset (\exists y)(y \in \hat{\omega} \wedge \langle x, y\rangle \in f))]\!]$$

(this is the translate of the formula which says that, for each $x \in \omega_2$, there is some $y \in \omega$ such that $f(x) = y$)

(1) We first show that formula (1) is true at p. It suffices to consider any a, b, c, and show that $p \vDash [\![((\langle a, b\rangle \in f \wedge \langle a, c\rangle \in f) \supset b = c)]\!]$. Suppose, on the contrary, that $p \nvDash [\![((\langle a, b\rangle \in f \wedge \langle a, c\rangle \in f) \supset b = c)]\!]$. Then, for some q with $p\mathcal{R}q$, $q \vDash [\![\langle a, b\rangle \in f]\!]$, $q \vDash [\![\langle a, c\rangle \in f]\!]$, and $q \vDash [\![\sim(b = c)]\!]$. Because $q \vDash [\![\langle a, b\rangle \in f]\!]$, it follows that for some g and some q' with $q\mathcal{R}q'$, $q' \vDash [\![\langle a, b\rangle \approx g\rangle$, $q' \vDash [\![\langle g \epsilon f\rangle]\!]$. Then it is easily seen that $g = \widehat{\langle n, \alpha\rangle}$ for some n and α. Because $q \vDash [\![\langle a, c\rangle \in f]\!]$ (and hence also $q' \vDash [\![a, c\rangle \in f]\!]$), similar reasoning will yield $q'' \vDash [\![\langle a, c\rangle \approx \widehat{\langle m, \beta\rangle}]\!]$ and $q'' \vDash [\![\widehat{\langle m, \beta\rangle}\epsilon f]\!]$ for some q'' with $q'\mathcal{R}q''$ and some m and β. From this it easily follows that $q'' \vDash [\![\hat{n} \approx \hat{m}]\!]$. This is because $(\forall x)(\forall y)((\langle x, y\rangle = \langle a, b\rangle) \supset x = a)$ is classically true in M, hence its translate is valid in the model. Then we can use this translate twice to get both $q'' \vDash [\![a \approx \hat{n}]\!]$ and $q'' \vDash [\![a \approx \hat{m}]\!]$, from which it follows that $q'' \vDash [\![\hat{m} = \hat{n}]\!]$. Similar reasoning shows that $q'' \vDash [\![\hat{\alpha} \approx b]\!]$, and $q'' \vDash [\![\hat{\beta} \approx c]\!]$. Then it follows (by Lemma 3.2 of Chapter 17) that n and m are the same set. Thus both $q'' \vDash [\![\widehat{\langle n, \alpha\rangle}\epsilon f]\!]$ and $q'' \vDash [\![\widehat{\langle n, \beta\rangle}\epsilon f]\!]$, which can easily be seen to imply $r \vDash \widehat{\langle n, \beta\rangle}\epsilon f$ and $s \vDash \widehat{\langle n, \alpha\rangle}\epsilon f$ for some r and s with $r\mathcal{R}s$. But, by the definition of the accessibility relation, s is an extension of r. Also, by the definition of f, $n \in \mathrm{Dom}(r)$ and $r(n) = \alpha$. Because s is an extension of r, and $n \in \mathrm{Dom}(s)$, it follows that $s(n) = \beta = \alpha$. Thus $\alpha = \beta$, meaning $q'' \vDash [\![\hat{\alpha} = \hat{\beta}]\!]$, meaning $q'' \vDash [\![\hat{\alpha} \approx \hat{\beta}]\!]$. This implies $q'' \vDash [\![b \approx c]\!]$, that is, $q'' \vDash [\![b = c]\!]$. But we also have $q \vDash [\![\sim(b = c)]\!]$ (by assumption), hence also $q'' \vDash [\![\sim(b = c)]\!]$, a contradiction.

(2) Now we move on to showing that formula (2) is true at p. It suffices to consider any a, b, c, and show that $p \vDash [\![((\langle a, c\rangle \in f \wedge \langle b, c\rangle \in f) \supset a = b)]\!]$. Suppose, on the contrary, that $p \nvDash [\![((\langle a, c\rangle \in f \wedge \langle b, c\rangle \in f) \supset a = b)]\!]$. Then, for some q with $p\mathcal{R}q$, $q \vDash [\![\langle a, c\rangle \in f]\!]$, $q \vDash [\![\langle b, c\rangle \in f]\!]$, and $q \vDash [\![\sim(a = b)]\!]$. Using near identical reasoning as in part (1) of this proof, it follows that for some r with $q\mathcal{R}r$ and some s with $r\mathcal{R}s$, $r \vDash \widehat{\langle n, \alpha\rangle}\epsilon f$ and $s \vDash \widehat{\langle m, \alpha\rangle}\epsilon f$ for some n, m, and α, where $r \vDash [\![a \approx \hat{n}]\!]$ and $r \vDash [\![b \approx \hat{m}]\!]$. Because $r \vDash \widehat{\langle n, \alpha\rangle}\epsilon f$, it follows, by the definition of f, that $n \in \mathrm{Dom}(r)$ and $r(n) = \alpha$. Similarly, $n \in \mathrm{Dom}(s)$ and $s(m) = \alpha$. But s is an extension of r, and because both r and s are 1-1 (all forcing conditions under consideration here are 1-1 functions), it follows that $n = m$. Hence $r \vDash [\![\hat{n} = \hat{m}]\!]$, implying $r \vDash [\![\hat{n} \approx \hat{m}]\!]$. Because $r \vDash [\![a \approx \hat{n}]\!]$ and $r \vDash [\![b \approx \hat{m}]\!]$, it then follows that $r \vDash [\![a \approx b]\!]$, that is, $r \vDash [\![a = b]\!]$. But $q \vDash [\![\sim(a = b)]\!]$, meaning $r \vDash [\![\sim(a = b)]\!]$, which is a contradiction.